Color Block Positioning Experiment

The main purpose of color block positioning is to achieve advanced functionality for color block following. The principle is to determine the distance and position information of the color block relative to the camera by calculating the center point coordinates of the color block in the camera's field of view, thereby achieving color block positioning. The experimental results show that it continuously finds and follows the center point of the color block.

1. Main Code

- Start Docker

After entering the Raspberry Pi 5 desktop, open the terminal and run the following command to start the corresponding Dofbot container:

```
./Docker_Ros.sh
```

Then, in the Docker terminal, input the following command to start jupyter-lab:

```
cd
jupyter-lab --allow-root
```

Then, in the browser, input the main board's IP address plus :9999:

```
IP:9999         // Example: 192.168.1.11:9999
```

Code path: /root/Dofbot_2D/src/dofbot_color_follow/Color block positioning.ipynb

- Import header files

```
import cv2 as cv
import threading
import random
from time import sleep
import ipywidgets as widgets
from IPython.display import display
from positioning import color_follow
```

- Main recognition function, simultaneously obtaining the target center point of the color block (color_x, color_y)

```
def follow_function(self, img, HSV_config):
        (color_lower, color_upper) = HSV_config
        self.img = cv.resize(img, (640, 480), )
        self.img = cv.GaussianBlur(self.img, (5, 5), 0)
        hsv = cv.cvtColor(self.img, cv.COLOR_BGR2HSV)
        mask = cv.inRange(hsv, color_lower, color_upper)
        mask = cv.erode(mask, None, iterations=2)
        mask = cv.dilate(mask, None, iterations=2)
        mask = cv.GaussianBlur(mask, (5, 5), 0)
        cnts = cv.findContours(mask.copy(), cv.RETR_EXTERNAL,
cv.CHAIN_APPROX_SIMPLE)[-2]
```

```
        if len(cnts) > 0:
            cnt = max(cnts, key=cv.contourArea)
            (color_x, color_y), color_radius = cv.minEnclosingCircle(cnt)

            if color_radius > 10:
                # Mark the detected color with the prototype coil
                # 将检测到的颜色用原形线圈标记出来
                cv.circle(self.img, (int(color_x), int(color_y)),
int(color_radius), (255, 0, 255), 3)
                print(color_x,color_y)
        return self.img
```

- Create controls

```
button_layout = widgets.Layout(width='200px', height='100px',
align_self='center')
# 输出控件 Output widget
output = widgets.Output()
# 颜色追踪 Color tracking
color_follow = widgets.Button(description='color_follow', button_style='success',
layout=button_layout)
# 选择颜色 Select color
choose_color = widgets.ToggleButtons(options=['red', 'green', 'blue', 'yellow'],
button_style='success',
             tooltips=['Description of slow', 'Description of regular',
'Description of fast'])
# 取消追踪 Cancel tracking
follow_cancel = widgets.Button(description='follow_cancel',
button_style='danger', layout=button_layout)

# 退出 exit
exit_button = widgets.Button(description='Exit', button_style='danger',
layout=button_layout)
# 图像控件 Image widget
imgbox = widgets.Image(format='jpg', height=480, width=640,
layout=widgets.Layout(align_self='auto'))
# 垂直布局 Vertical layout
img_box = widgets.VBox([imgbox, choose_color],
layout=widgets.Layout(align_self='auto'))
# 垂直布局 Vertical layout
Slider_box = widgets.VBox([color_follow,follow_cancel,exit_button],
                          layout=widgets.Layout(align_self='auto'))
# 水平布局 Horizontal layout
controls_box = widgets.HBox([img_box, Slider_box],
layout=widgets.Layout(align_self='auto'))
# ['auto', 'flex-start', 'flex-end', 'center', 'baseline', 'stretch', 'inherit',
'initial', 'unset']
```

Main process:

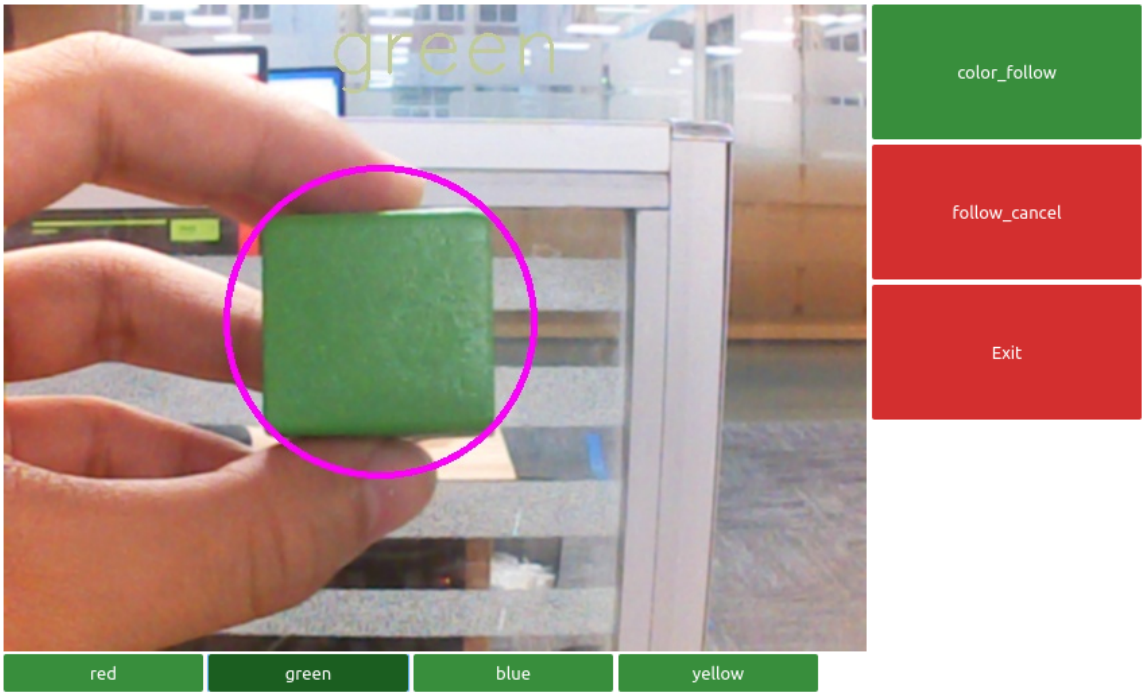
```
def camera():

    global HSV_learning,model
    # 打开摄像头 Open camera
    capture = cv.VideoCapture(0)
```

```
    capture.set(3, 640)
    capture.set(4, 480)
    capture.set(5, 30)
    # Be executed in loop when the camera is opened normally
    # 当摄像头正常打开的情况下循环执行
    while capture.isOpened():
        try:
            _, img = capture.read()
            img = cv.resize(img, (640, 480))

            if model == 'color_follow':
                img = follow.follow_function(img, color_hsv[choose_color.value])
                cv.putText(img, choose_color.value, (int(img.shape[0] / 2), 50),
cv.FONT_HERSHEY_SIMPLEX, 2, color[random.randint(0, 254)], 2)

            if model == 'learning_color':
                img,HSV_learning = follow.get_hsv(img)

            if model == 'Exit':
                cv.destroyAllWindows()
                capture.release()
                break
            imgbox.value = cv.imencode('.jpg', img)[1].tobytes()
        except KeyboardInterrupt:capture.release()
```

When the program runs to the end, click color_follow, and you can see it will follow the center coordinates of the color block for selection.